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ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

UNIT: COMPUTER SOFTWARE

ISCED UNIT CODE: 0619 451 06A

CONTINUOUS ASSESSMENT TEST (CAT) 3

ELEMENT: SOFTWARE MAINTENANCE, MONITORING, UPGRADES AND SAFETY PROCEDURES

TIME: 1½ HOURS

TOTAL MARKS: ~~50~~ 60

CANDIDATE NAME: KHIMAI MARCUS

REGISTRATION NUMBER: ______________________

DATE: 30/6/2026

INSTRUCTIONS TO CANDIDATES

1. Answer ALL questions.
2. Answer sheets will be provided separately.
3. Read each question carefully before answering.
4. Marks for each question are indicated in brackets.
5. The total marks for this paper are 50.

1. Define the term software maintenance. (2 Marks)
2. State two importance of software maintenance. (2 Marks)
3. List two activities that may be included in a software maintenance schedule. (2 Marks)
4. Differentiate between:

   a) Corrective maintenance

   b) Preventive maintenance (4 Marks)

5. Match the following software maintenance types with their descriptions: (4 Marks)

   a) Adaptive Maintenance

   b) Perfective Maintenance

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c) Preventive Maintenance

d) Corrective Maintenance

Descriptions:

i. Improving software performance and usability Perfective

ii. Correcting software faults and errors Corrective

iii. Preventing future software failures Preventive

iv. Modifying software to operate in a new environment Adaptive

6. State any three types of software updates. (3 Marks)
7. State three Personal Protective Equipment (PPE) used when handling computer equipment. (3 Marks)
8. Explain four importance of software upgrades. (4 Marks)
9. Describe three types of software upgrades. (6 Marks)
10. Define software functionality monitoring. (2 Marks)
11. Explain the following types of operating system event logs: (8 Marks)

a) Error Event Logs

b) Warning Event Logs

c) Information Event Logs

d) Success Audit Event Logs

12. Prepare a simple monthly software maintenance schedule for a computer laboratory showing four maintenance activities. (4 Marks)
13. Explain three importance of software updates. (6 Marks)
14. Explain three safety measures observed during software maintenance activities. (6 Marks)
15. Explain the importance of proper use of tools and equipment when carrying out software maintenance. (4 Marks)

140
14
56

1. Software maintenance is the process of modifying, updating and improving a software application after its initial launch. 2

2. Enhance security of software
Boosts performance of the operating system 2

3. Security patching 2
Bug fixing

4. Corrective maintenance reacts to fix existing problems while preventive maintenance acts to stop future problems 2

5. Improving software performance and usability → Perfective maintenance
Correcting software faults and errors. corrective maintenance 3
Preventing future software failures → Preventive maintenance.
Modifying software to operate in a new environment. → Adaptive maintenance

6. Security Patches ?, updates
Bug fixes ?,
~~Fe~~ Feature upgrades.

7. Dust mask
Safety boots. 3
Safety glasses.

8 Fixes system bugs → ~~improv~~ Eliminate glitches and software crashes improving overall
Patches security flaws → Fixes any available vulnerability preventing hackers & preventing breaches. 4
Adds new features → Introduces ~~ne~~ modern capabilities user ~~ma~~ interface enhance experience.
Boost hardware performance → Optimizes code execution speeds, which reduce drain and system lag.

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13 Q. Security upgrades → Helps in security risk management
Version upgrade/update → New version bring much better user interfaces and experience. 8
Utility Program update → By updating the utilities they create a much better user environment through better computer system management.

10. Software functionality monitoring is continuous process of keep track of software to ensure it meets end users expectations. 8

11. An Error event log indicate immediate problem ~~caused by~~ causing specific application or component fail. 8
Warning event log → this indicates a not immediate but potential future problem
Success Audit Event Log → is a security focused record showing system access attempt action completed successfully. 8

8

12.

Week	activity	Description.
Week 1	Operating system and application Patching	Run automated updates for the operating system and core software
Week 2	Security scanning	Execute full malware scans and update antivirus definitions.
Week 3	Storage cleanup	Clear temporary files, browser caches and empty recycle bins.
Week 4	System Backup	Create full system restore Points and critical laboratory data.

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9. Security upgrades. Closes security vulnerabilities by defending system against malware and cyberattacks.
Fixes code defects → this by eliminating persistent system bugs.
Optimizes system speed → Boost processing efficiency leading to faster load times.
Introduces new features → delivers upgraded user interfaces

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16

14 Perform full backup Create ~~complete~~ system restore point and data backups befor modifying any software.

Schedule scheduled downtime. → execute ~~major~~ maintenance during off-peak hours to prevent active user disruption

Enforce Access control Restrict ~~maintenance~~ permission exclusively to authorized administrators using secure authentication

15 Prevent system crashes by using write tools ~~that~~ human errors are eliminated.

Accelerates troubleshooting diagnostics and by analysis tools Pinpoint exact root cause of software bugs.

Maximize resource efficiency Automation enable maintenance of hundreds of machines simultaneously.

Secure critical data. Use of proper tools ensure user file are not corrupted.

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